METRIC

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PERFORMANCE SPECIFICATION

AIR CLEANER ASSEMBLY

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 425 liter per second (L/s) [900 cubic foot per minute (ft^3/min)], fender-mounted, heavy-duty, two-stage, dry-type air cleaner assembly referred to herein as the "assembly" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2940

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-F-62309 - Filter Element, Air.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

DEPARTMENT OF DEFENSE

MIL-STD-193 - Painting Procedures and Marking for Vehicles, Construction Equipment, Material Handling Equipment, and Spare Parts.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11669740 - Filter Element.
12325888 - Air Cleaner Assembly.

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(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, Ml 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4	- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005,611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A123	- Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel (DoD Adopted).
ASTM C800	- Standard Specification for Glass Fiber Blanket Insulation (Aircraft Type) (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

GLOBAL ENGINEERING

GM 9540P	- Accelerated Corrosion Test.

(Application for copies should be addressed to Global Engineering, 15 Invernessway. Englewood. CO 80112.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE AS 478	- identification Marking Methods (DoD Adopted).

{Applications for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4. First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent assemblies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be uniform and free from imperfections or defects which affect their serviceability y. All nonmetallic materials shall be inherently fungistatic or treated to resist fungus growth except when used for components of potted or otherwise moisture sealed assemblies (see 4.7. 1).

3.2.1 Corrosion protection.. The components and assembly shall provide corrosion protection equal to or exceeding that provided by hot dip galvanized 1020 steel, with coating thickness in accordance with ASTM Al23 (or minimum coating thickness of 19 micrometers (0.75 mil) on pre-galvanized sheet, 0.16 millimeters (mm) (0.063 inch (in)) or less, with zinc phosphate pre-treatment. A proposed design shall be compared to a galvanized sample (as described above) using the GM 9540P, Method B, 120 cycles, or until prior failure of one of the items with defects such as extensive corrosion at scribe or significant penetration of base material (see 4.7.1).

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cows (see 6.3).

3.3 Design and construction.

3.3.1 Configuration. The assembly shall be constructed as specified herein and in accordance with Drawing 12325888. The assembly shall incorporate a precleaner, filter element and a vehicle exhaust dust ejection system (VEDES) precleaned scavenging manifold (see 4.7.1, 4.7.2 and 6.2).

3.3.1.1 Filter element. The filter element shall be in accordance with MIL-F-62309 and Drawing 11669740 (see 4.7.1)

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3.3.2 Serviceability. Servicing, cleaning, and reassembly shall be accomplished without the use of special tools and without disturbing the air cleaner connections to the engine (see 4.7.1).

3.3.3 Weight. The weight of the assembly shall not exceed 272 kilogram (kg) [600 pounds (lbs)] (see 4.7.3).

3.3.4 Interchangeability. The assembly shall be designed and fabricated such that two or more items possessing functional and physical characteristics which are equivalent in performance and durability are capable of being exchanged one for the other without alteration of the items themselves or of adjoining items except for adjustment, and without selection for fit, repair parts, or performance (see 4.7.1).

3.4 Performance. Except as otherwise specified herein, performance requirements shall be met under the following conditions (see 4.7.4):

a. Temperature: 23 ± 10 degrees Celsius (°C) [73 ± 18 degrees Fahrenheit (°F)].
b. Humidity: 35 ± 5 percent (%).
c. Density of air: 1.17 kilogram per cubic meter (kg/m^3) [0.073 pound per cubic foot (lb/ft^3)] at 27°C (80°F) and 101 kilopascal (kPa) [30 inches (in) of mercury (Hg)].
d. Pressure: 101 ± 3 kPa (30 ± 1 in of Hg).
e. Dust: Chemical analysis of the test dust shall be as shown in table I. Particle size distribution shall be as shown in table II.
f. Dust feed rate: 0.88 ± 0.18 gram per cubic meter (g/m^3) [0.025 ± 0.005 gram per cubic foot (g/ft^3)] of air.

TABLE I. Chemical analysis of test dust.

Substance	Percent of weight
Silicon dioxide ($Si0_2$)	67 to 69
Ferric oxide (Fe_20_3)	3 to 5
Aluminum oxide (Al_20_3)	15 to 17
Calcium oxide (Ca0)	2 to 4
Magnesium oxide (Mg0)	0.5 to 1.5
Total alkalis as sodium oxide (Na_20)	3 to 5
Ignition loss	2 to 3

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TABLE II. Particle size distribution.

Particle size (micrometers)	Percent of total weight	
	Coarse grade	Fine grade
0 - 5	12 ± 2	39 ± 2
5 - 10	12 ± 3	18 ± 3
10 - 20	14 ± 3	16 ± 3
20 - 40	23 ± 3	18 ± 3
40 - 80	30 ± 3	9 ± 3
80 - 200	9 ± 3	----

3.4.1 Initial airflow restriction. The initial airflow restriction of the assembly shall not exceed 2.2 kPa (9 in of water) at a rated flow of 425 L/s [900 ft^3/min] (see 4.7.4.1)

3.4.2 Dust capacity. When operating with a stepped airflow in accordance with table III and fed coarse dust in accordance with table II, the assembly shall be capable of operating to an airflow restriction of 5 kPa (20 in of water). The time for the assembly to reach the restriction of 5 kPa shall be not less than 80 hours. The coarse dust efficiency shall be not less than 99.9% and the scavenge system efficiency shall be not less than 97% (see 4.7.4.2).

TABLE III. Stepped airflow.

Time (minutes)	Rated airflow (%)	Rated airflow L/s	Rated airflow (ft^3/min)	Scavenge airflow L/s	Scavenge airflow (ft^3/min)
10	100	425	(900)	33	(70)
10	80	340	(720)	43.4	(92)
10	60	255	(540)	50.5	(107)
10	40	170	(360)	56.6	(120)
10	20	85	(180)	60.4	(128)

3.4.3 Dust efficiency. The assembly dust filtration efficiency shall be not less than 99.5% when subjected to an airflow rate of 425 L/s (900 ft^3/min), a scavenge airflow rate of 33 L/s (70 ft^3/min), and fed fine dust in accordance with table 11 for 30 minutes (see 4.7.4.3).

3.4.4 Post service airflow restriction. After cleaning the filter element, the assembly airflow restriction shall not exceed the recorded initial airflow restriction (see 3.4.1) by more than 0.25 kPa (1 in of water) at a rated airflow of 425 L/s (900 ft^3/min) (see 4.7.4.4).

3.4.5 Air leakage resistance. There shall be no more than a 0.5 kPa (2 in of water) drop in 3 minutes when the assembly is subjected to a vacuum of 7.5 ± 0.1 kPa (30 ± 0.5 in of water) (see 4.7.4.5).

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3.4.6 Vibration. The assembly shall be capable of withstanding the vibration profile of figure 1 for a period of 80 minutes (including forced dwells of 13.33 minutes each) in each of the three mutually perpendicular axes (see 4.7.4.6).

Axis	Sweep acceleration 5 to 500 to 5 Hz 1/ at 0.9 octaves/minute	Forced dwells					
	G-level 2/	Primary		2nd		3rd	
		Accel. (G-level)	Freq. (Hz)	Accel. (G-level)	Freq. (Hz)	Accel. (G-level)	Freq. (Hz)
Vertical (Z)	0.9	4.2	20	2.4	105	2.4	265
Transverse (Y)	0.9	1.5	50	1.8	100	-	-
Longitudinal (X)	0.9	1.2	245	-	-	-	-

1/ Hertz (Hz). 2/ Gravity (G).

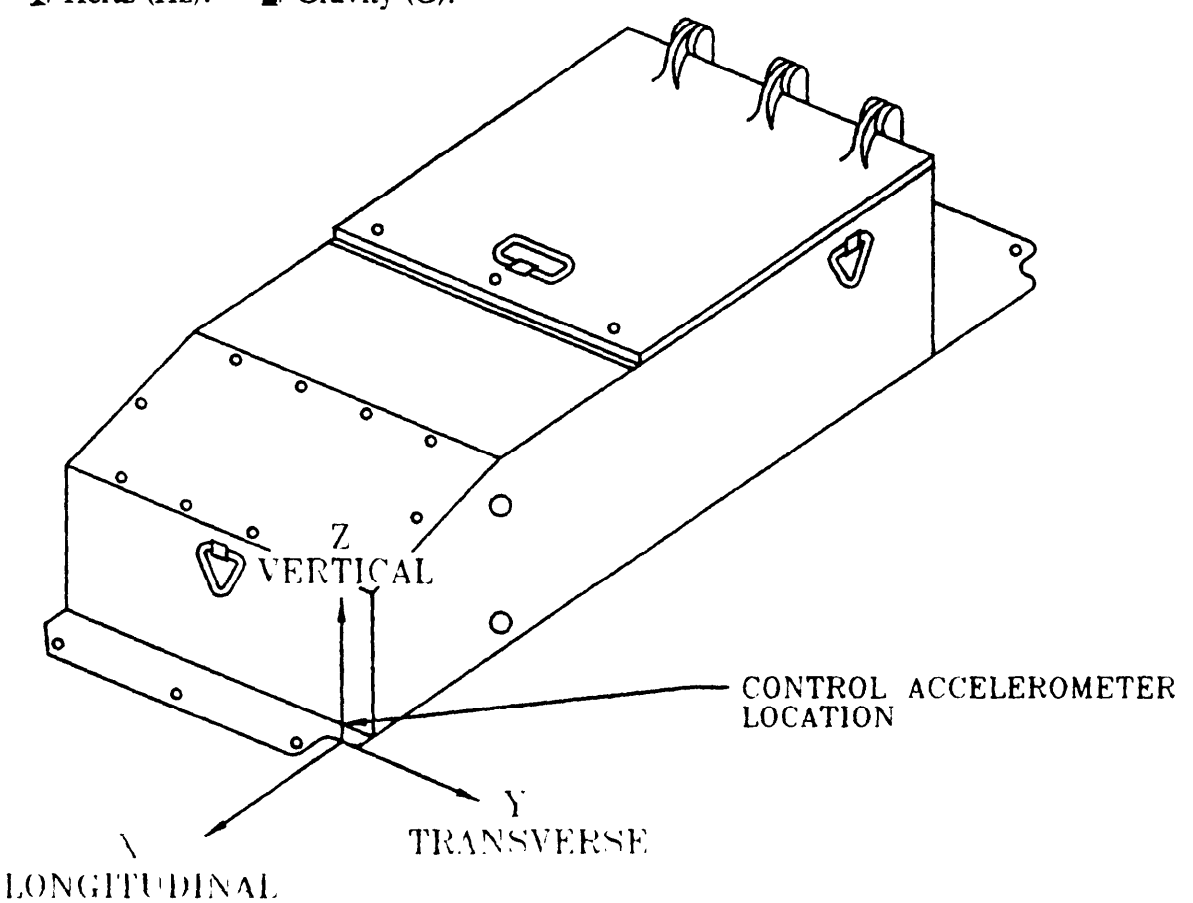


FIGURE 1. Vibration levels for assembly.

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3.5 Painting. Unless otherwise specified (see 6.2), the preparation, conditioning, and painting of the assembly shall be in accordance with applicable provisions of MIL-STD-193. Exterior color shall be in accordance with color chip 24087 (olive drab) of FED-STD-595 (see 4.7.2).

3.6 Identification marking. Identification and marking shall be permanent and legible in accordance with SAE AS 478 and shall include the maximum rated airflow (see 4.7.2).

3.7 Workmanship. The assembly shall be cleaned of any foreign material which might detract from the intended operation, function, or appearance of the assembly. Screws, nuts, and bolts shall show no evidence of cross threadings, mutilation, or detrimental or hazardous burrs, and shall be firmly secured. Bearing assemblies shall be free of rust, discoloration, and imperfections of ground, honed, or lapped surfaces. Contacting surfaces shall be free of tool marks, gouge marks, nicks or other surface-type defects. There shall be no detrimental interference, binding, or galling (see 4.7.2).

4. VERIFICATION

4.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision, maintenance, and accuracy of all inspection equipment necessary to assure that supplies and services conform to contract requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
    1. Preproduction inspection (see 4.4.1).
    2. Initial production inspection (see 4.4.2).
b. Conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under conditions specified in 3.4a, b, c and d.

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein.

4.4.1 Preproduction inspection. Unless otherwise specified (see 6.2), the preproduction sample shall consist of two assemblies. Preproduction inspection shall consist of inspection as specified in table IV, in the order listed in table V.

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4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two assemblies from the first 10 assemblies, produced under the production contract, for initial production inspection. Initial production units shall be inspected as specified in table IV, and in the order listed in table V.

4.4.3 First article inspection failure. Test item deficiencies, during or as a result of the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found, during or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE IV.Classification of inspections.

Title	Requirement	Inspec-tion	First article	Conformance inspection	
				Examination	Test
Materials and construction	3.2 thru 3.3.2 and 3.3.4	4.7.1	X		
Defects (see 4.5.2)	3.3.1, 3.5, 3.6, and 3.7	4.7.2	X	X	
Weight	3.3.3	4.7.3	X		X
Performance	3.4	4.7.4			
Initial airflow restriction	3.4.1	4.7.4.1	X		X
Dust capacity	3.4.2	4.7.4.2	X		
Dust efficiency	3.4.3	4.7.4.3	X		
Post service air-flow restriction	3.4.4	4.7.4.4	X		
Air leakage resistance	3.4.5	4.7.4.5	X		X
Vibration	3.4.6	4.7.4.6	X		

TABLE V. Sequence of tests.

Test sequence	Inspection	First article sample	
		1	2
Materials and construction	4.7.1 and 4.7.2	X	X
Weight	4.7.3	X	X
Air leakage resistance	4.7.4.5	X	X
Initial airflow restriction	4.7.4.1	X	X
Dust capacity	4.7.4.2	X	X
Post service airflow restriction	4.7.4.4 1/	X	X
Dust efficiency	4.7.4.3	X	X
Vibration	4.7.4.6	X	X

1/ See 4.7.4g.

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4.5 Conformance inspection. Conformance inspection shall include the examinations of 4.5.2 and the tests of 4.5.3.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling. Samples for conformance examination shall be selected in accordance with ASQC ZI.4.

4.5.2 Examinations. Each sample selected in accordance with 4.5.1.2 shall be examined as specified in table VI.

TABLE VI. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
103	Identification marking, improper (see 3.6).	Visual
104	Faulty workmanship affecting performance (see 3.7).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE
202	Paint, improper application or color (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Samples selected in accordance with 4.5.1.2 shall be subjected to the tests specified in table IV.

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4.6 Failure. Failure of any assembly to pass any of the specified conformance inspections shall because for the contracting activity to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the contracting activity.

4.7 Methods of ispection.

4.7.1 Materials and construction. Conformance to 3.2, 3.2.1, 3.3.1, 3.3.1.1, 3.3.2, and 3.3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3.1 3.3.2, 3.5, 3.6, and 3.7 shall be determined by examination for the defects listed in table VI. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7.3 Weight. To determine conformance to 3.3.3, the assembly shall be weighed.

4.7.4 Performance. Except as otherwise specified herein, test conditions and procedures shall be in accordance with 3.4 and the following:

a. The test setup for all tests requiring airflow shall be as shown in figure 2.
b. Dust feed rates shall be averaged over 10-minute intervals.
c. Measured airflow restriction values shall be corrected to the specified air density (see 3.4c) to an accuracy of 25 Pa (0.1 in of water).
d. Actual airflow values used in tests shall be corrected to standard conditions (see 3.4c) and shall be within ± 2% of specified values.
e. The assembly shall be mounted in its normal operating attitude as shown in figure 2 and shall be electrically grounded.
f. In all tests, the assembly shall be connected to a turbine vacuum source as shown in figure 2. The vacuum source shall provide a scavenge airflow as specified in 3.4.2.
g. For first article testing, two filter elements shall be used for each assembly tested. The second filter shall be inserted subsequent to the post service airflow restriction test prior to the dust efficiency test.
h. Air shall be drawn through the assembly with a turbine vacuum source to insure filter element dryness before starting tests.

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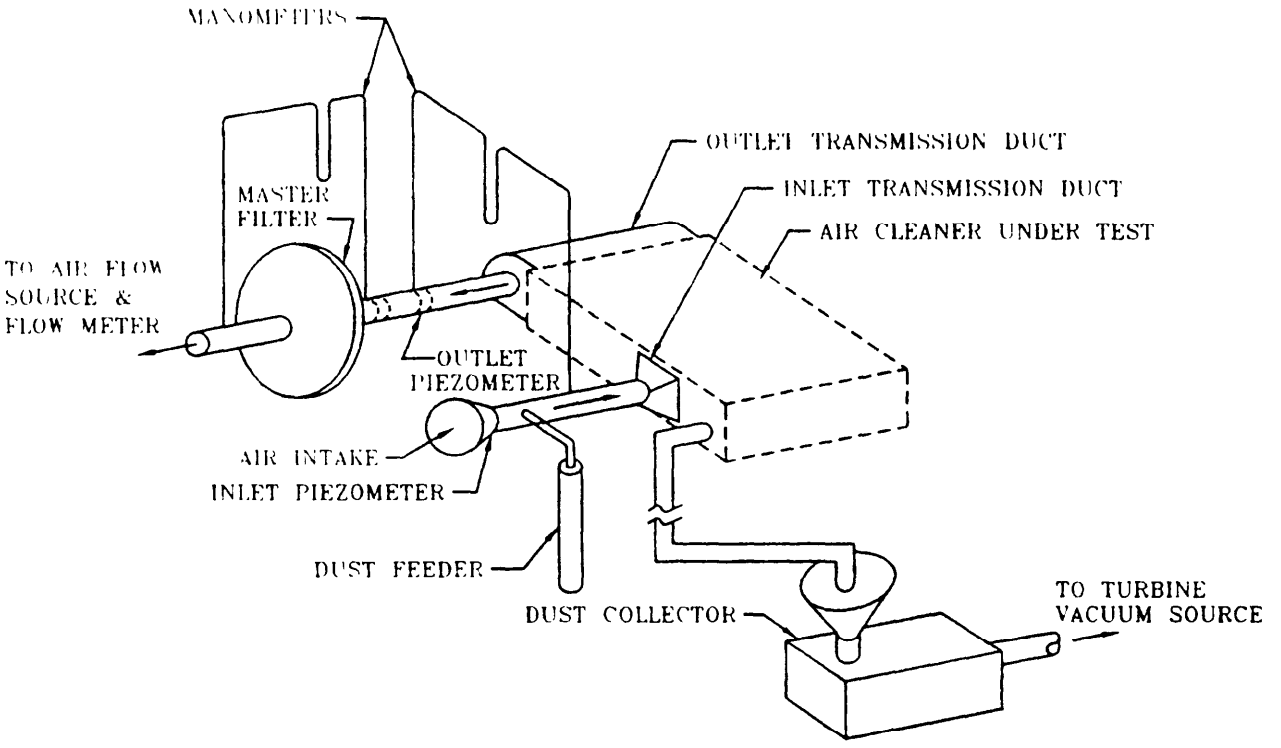


FIGURE 2. Typical test set up (air cleaner with VEDES scavenging.

4.7.4.1 Initial airflow restriction. To determine conformance to 3.4.1, the assembly shall be subjected to a dust free airflow of 425 L/s (900 ft^3/min) through a new filter element. The restriction to flow shall be measured and recorded. The humidity condition of 3.4 shall not apply.

4.7.4.2 Dust capacity. To determine conformance to 3.4.2, the assembly shall be mounted in the test set up of figure 2, and 1 hour cycles with stepped flow rates as specified in table III shall be completed. The restriction to flow (pressure drop through the assembly and master filter) shall be recorded each time the flow is adjusted to 425 L/s (900 ft^3/min). The test shall be conducted until the airflow restriction at 425 L/s (900 ft^3/min) reaches 5.0 kPa (20 in of water). Coarse dust shall be fed into the input side of the assembly at 0.88 $\pm$ 0.18 g/m^3 (0.025 $\pm 0.005\ g/ft^3$) of actual input airflow. The operational time shall be at least 80 hours. At approximately 20-hour intervals of continuous operation or whenever the test setup is required to be shutdown, the filter element shall be viewed through the assembly outlet for visual signs of dust tracking on the clean side of the seal or the filter element media. Any visual signs of dust tracking shall be cause for rejection. At the conclusion of testing, the contractor shall verify that the coarse dust efficiency of 99.9% was attained and that the scavenge system efficiency was not less than 97%. The following formulas shall be used to calculate coarse dust or scavenge system efficiency:

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$$\text{Scavenge system efficiency (\%)} = \frac{W_1 - [\,(W_4 - W_5) + (W_2 - W_3)\,]}{W_1} \times 100$$

$$\text{Coarse dust efficiency (\%)} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where: W_1 = Weight of coarse dust fed into the assembly.

W_2 = Weight of master filter element and its entrapped dust.

W_3 = Weight of master filter element at start of test.

W_4 = Weight of filter element, its entrapped dust and dust encasing the sides of the filter housing compartment.

W_5 = Weight of filter element at start of test.

4.7.4.3 Dust efficiency. To determine conformance to 3.4.3, the assembly shall be mounted in the test setup of figure 2 with a master filter (see 4.7.4.3.1) at the output end of the assembly. Fine dust as specified in table II shall be placed in a dust feeder and shall be fed into the input side of the assembly at 0.88 ± 0.18 /m^3 (0.025 ± 0.005 g/ft^3) of actual input airflow for 30 minutes with an airflow of 425 L/s (900 ft /min) and a scavenge airflow of 33.0 L/s (70 ft^3/min). Restriction at the start of the test and at 10 minute intervals throughout the test shall be recorded. The master filter shall be removed from the test setup taking care not to disturb the entrapped dust and reweighed as specified in 4.7.4.3.1. All fine dust and master filter element weighings shall be accurate to 0.01 gram. Other weighings shall be accurate to 0.01 gram wherever possible but never less accurate than 0.1 gram. The dust efficiency of the assembly shall be 99.5% when computed using the following formula:

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$$\text{Dust efficiency (\%)} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where: W_1 = Weight of dust fed into the assembly.

W_2 = Weight of master filter and its entrapped dust.

W_3 = Weight of master filter before test.

4.7.4.3.1 Preparation of the master filter. A master filter conforming to ASTM C800 shall be placed in an air oven and dried for 6 hours at a temperature of 93°C to 116°C (200°F to 240°F), prior to insertion in the test setup. This drying time may be reduced, and the master filter considered as completely dry, if there is no change in weight of the master filter over two consecutive 10-minute internals.

4.7.4.4 Post service airflow restriction. To determine conformance to 3.4.4, after the dust capacity test of 4.7.4.2, the filter element shall be removed and cleaned of dust by using clean, dry, compressed air with 690 ± 69 kPa [100 ± 10 pound per square inch (lb/in^2)] line pressure, and a nozzle inside diameter of not less than 3.18 mm (0.13 in). The air nozzle shall be inserted into each pocket or serration of the filter element using a back and forth motion along the length of each serration and a direct flow into each pocket from inside out. After the filter element is thoroughly air cleaned, it shall be reassembled in the assembly. The assembly shall be subjected to a dust free flow of 425 L/s (900 ft^3/min) of air through the filter element. The restriction to flow shall be measured and recorded.

4.7.4.5 Air leakage resistance. To determine conformance to 3.4.5, the assembly shall be sealed at all inlets and outlets and subjected to a vacuum of 7.5 ± 0. kPa (30 ± 0.5 in of water). The vacuum decay (leakage) shall be monitored for 3 minutes. The humidity condition of 3.4 shall not apply.

4.7.4.6 Vibration. To determine conformance to 3.4.6, the assembly shall be mounted on vibrating equipment with the filter element inserted and scavenge exhaust attached (see figure 2). The assembly shall vibrate for a period of 80 minutes at an amplitude of 0.9 gravity units (g) and a frequency rate from 5 to 500 to 5 hertz (Hz) in each of the three mutually perpendicular axes as shown in figure 1. The forced dwells of figure 1 shall be performed in place of the resonance search and dwells. Connections and instrumentation shall be attached to permit testing as follows:

a. Test level shall be as specified in figure 3 when conducted at ambient temperatures. The time schedule shall be a total of 80 minutes per axis including the forced dwells of figure 1.

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b. The test item shall be vibrated along each axis in accordance with the condition above. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 4. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown in figure 4 for the specified range. Linear sweep rates shall be substituted for the logarithmic sweep rate. When linear sweep rates are used, the total frequency range shall be divided into logarithmic frequency bands having similar time intervals such that each time interval is the time of ascending plus a descending sweep for the corresponding band. The sum of these time intervals shall equal the sweep time specified for the applicable frequency range. The linear sweep rate for each band is then determined by dividing each bandwidth in Hz by one half the sweep time in minutes for each band. The logarithmic frequency bands may be readily determined from figure 4. The frequency bands and linear sweep rates shown in table VII shall be used for the 2 (or 5) to 500 Hz and 5 to 2000 Hz frequency ranges. For test frequency ranges of 100 Hz or less, no correction of the linear sweep rate is required.

c. After vibration, the assembly shall be examined for structural damage and shall be subjected to an air leakage test as specified in 4.7.4.5 and the dust efficiency test as specified in 4.7.4.3. The filter element shall be cleaned as specified in 4.7.4.4 prior to performing the dust efficiency test.

TABLE VII. Linear cycling rates.

Total frequency range (Hz)	Frequency band (Hz)	Sweep time in minutes (min-max-min)	Linear cycling rate (Hz/min)
2 - 500 or 5 - 500 as applicable	2 to 5 5 to 22.5 22.5 to 110 110 to 500	3 6 5 4	2 5.8 35 195
5 - 2000	5 to 22.5 22.5 to 110 110 to 500 500 to 900 900 to 2000	6 5 4 3 2	5.8 35 195 267 1100

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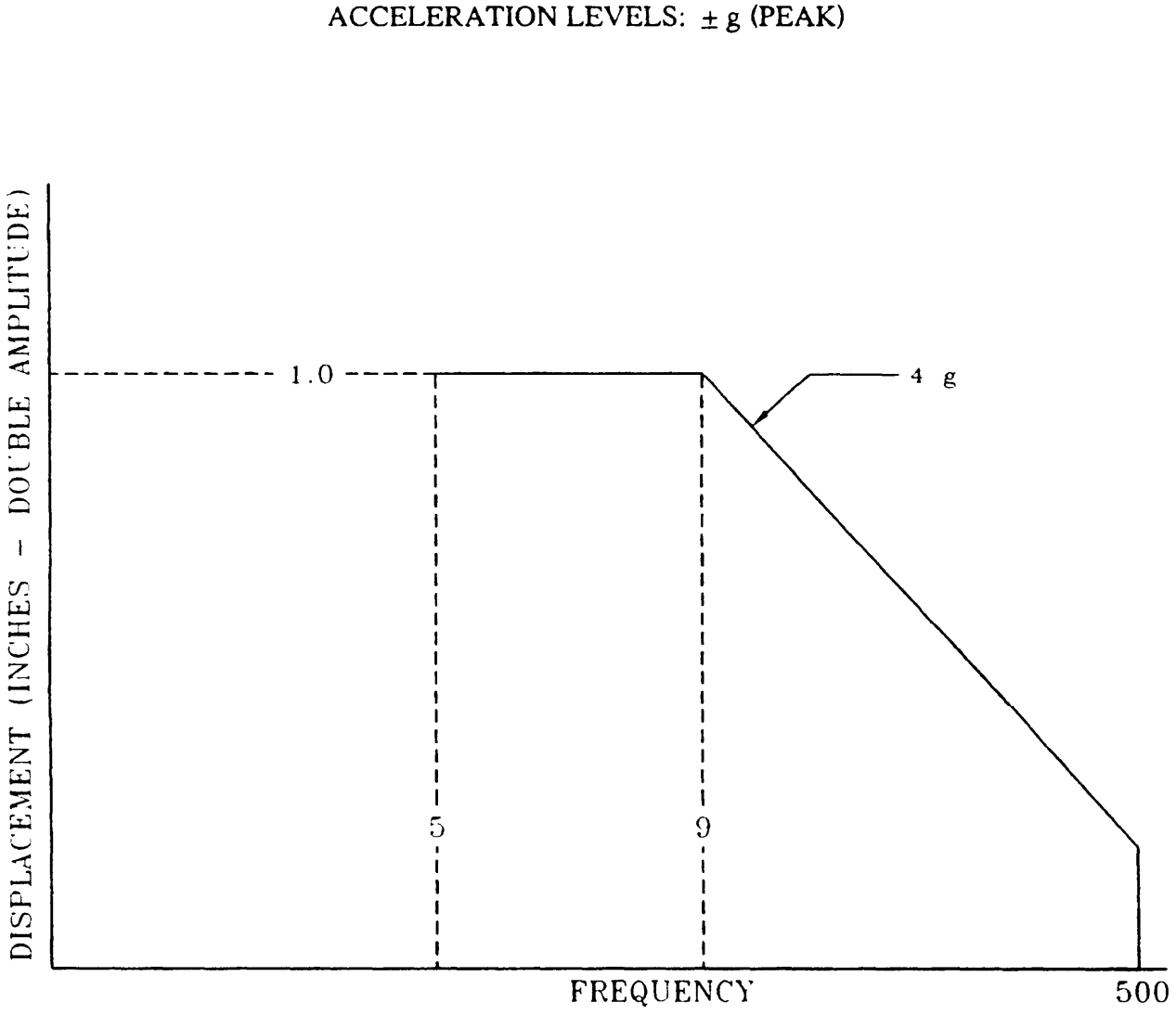


FIGURE 3. Vibration test curve for equipment installed in ground vehicles.

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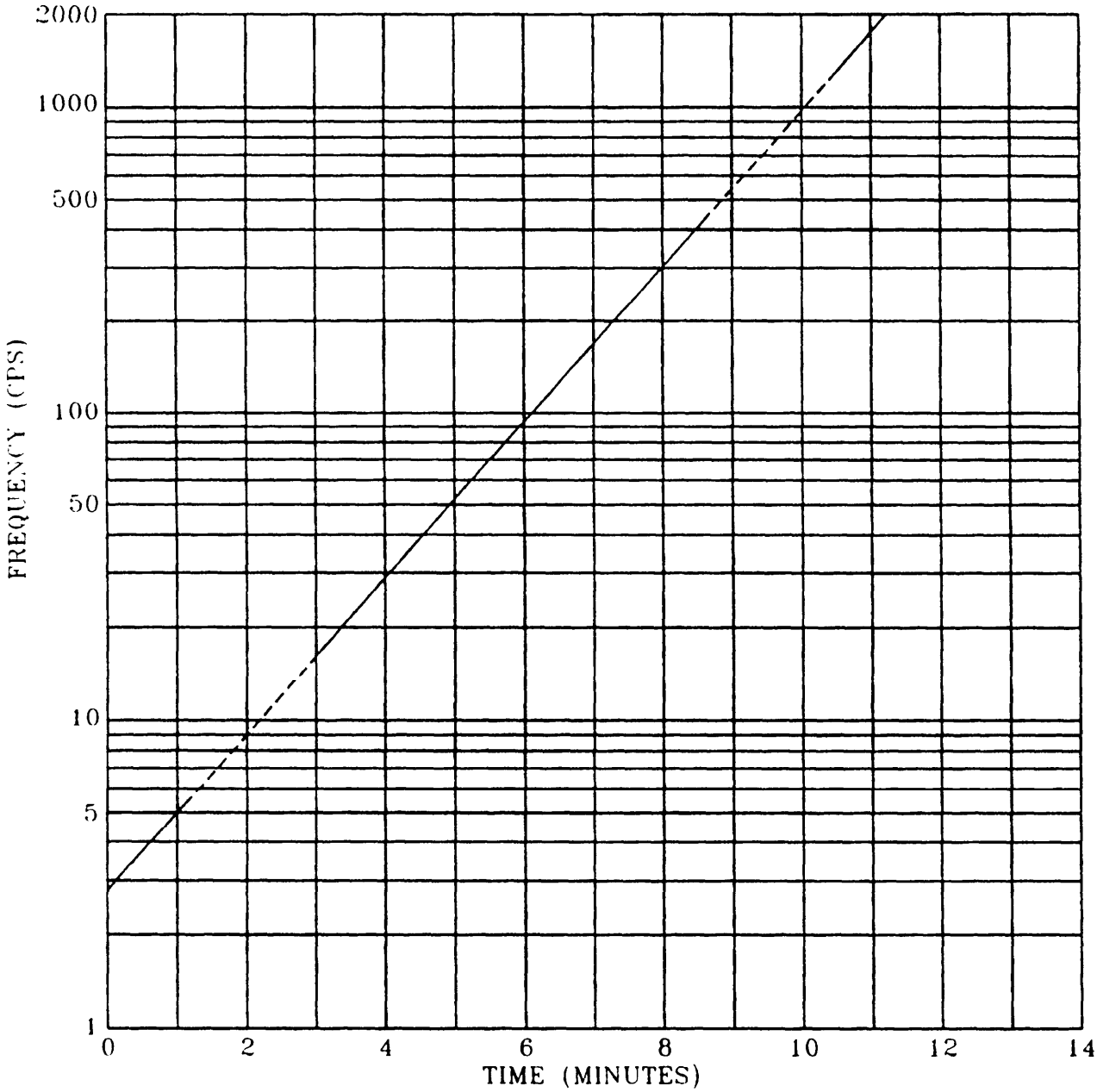


FIGURE 4. Logarithmic sweep.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Intended use. The filter element is intended specifically for use on full-tracked, combat M60 series tanks.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article inspection is required (see 3.1).
d. Title, number, and date of the applicable drawing (see 3.3.1).
e. If painting should be other than as specified (see 3.5).
f. If responsibility for inspection equipment should be other than as specified (see 4.1).
g. If inspection conditions should be other than as specified (see 4.3).
h. If preproduction sample should be other than as specified (see 4.4.1).
i. If initial production sample should be other than as specified (see 4.4.2).
j. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

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6.3.2 Solid waste. “Solid waste" is defined as (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved materials in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under sections 402 of the Clean Air Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954042 U. S. C.201 1 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Combat vehicle
Dry-type
Fender-mounted
Filter element
Full-tracked
Heavy-duty
M60 series
Tanks
Two-stage

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing activity:
Army -AT

(Project 2940-0153)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-62337A(AT)	960219

3. DOCUMENT TITLE: Air Cleaner Assembly

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (810) 574-8745 (2) AUTOVON: 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-F/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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